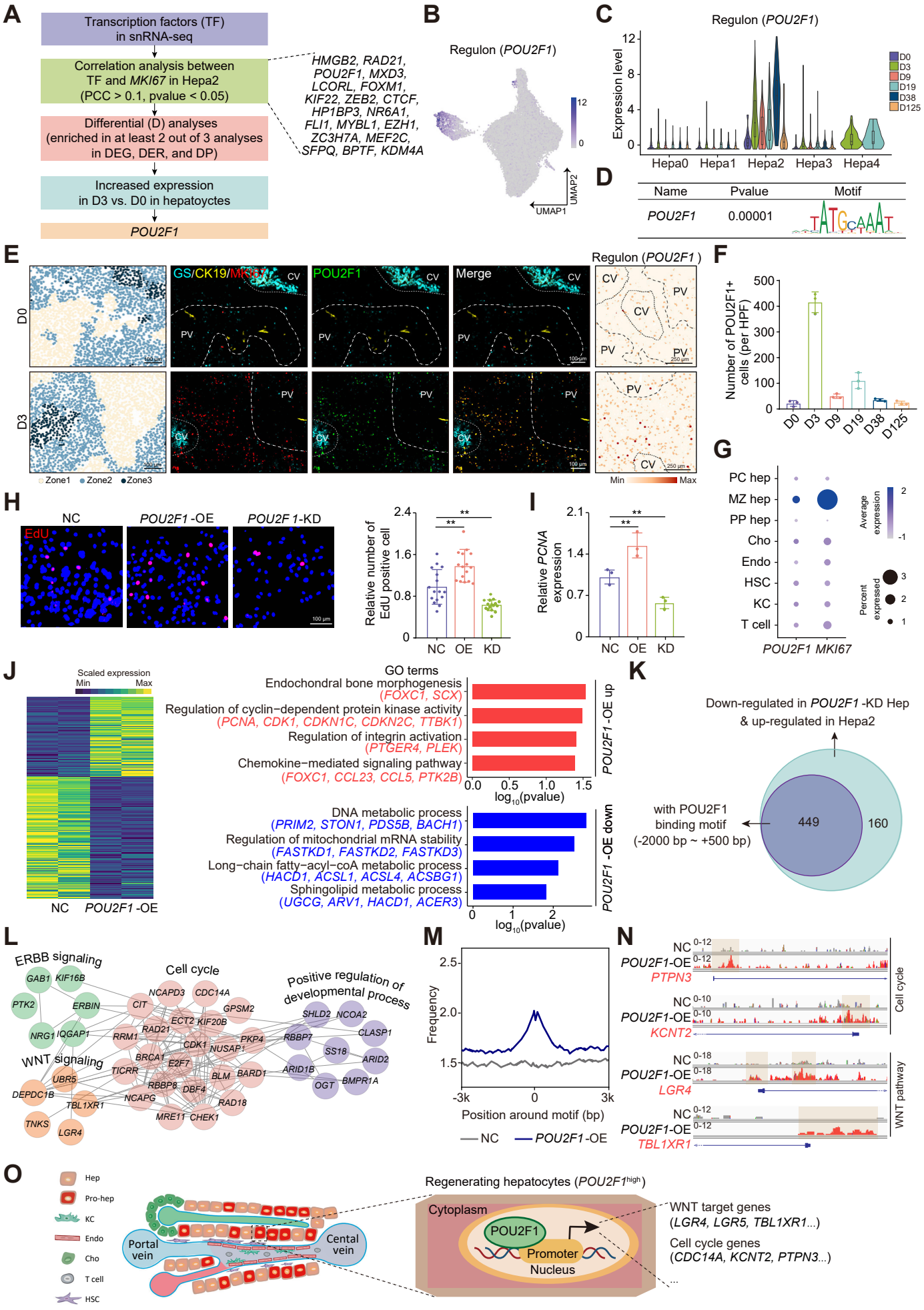

A
Transcription factors (TF) in snRNA-seq
Correlation analysis between TF and MKI67 in Hepa2 (PCC > 0.1, pvalue < 0.05)
Differential (D) analyses (enriched in at least 2 out of 3 analyses in DEG, DER, and DP)
Increased expression in D3 vs. D0 in hepatoyctes
POU2F1
HMGB2, RAD21, POU2F1, MXD3, LCORL, FOXM1, KIF22, ZEB2, CTCF, HP1BP3, NR6A1, FLI1, MYBL1, EZH1, ZC3H7A, MEF2C, SFPQ, BPTF, KDM4A
B
Regulon (POU2F1)
UMAP1
UMAP2
C
Regulon (POU2F1)
Expression level
Hepa0 Hepa1 Hepa2 Hepa3 Hepa4
D0 D3 D9 D19 D38 D125
D
Name Pvalue Motif
POU2F1 0.00001
E
D0
D3
GS/CK19/MKI67
POU2F1
Merge
Regulon (POU2F1)
CV
PV
Zone1 Zone2 Zone3
Min Max
F
Number of POU2F1+ cells (per HPF)
D0 D3 D9 D19 D38 D125
G
PC hep
MZ hep
PP hep
Cho
Endo
HSC
KC
T cell
POU2F1 MKI67
Average expression
Percent expressed
H
NC
POU2F1 -OE
POU2F1 -KD
EdU
Relative number of EdU positive cell
NC OE KD
I
Relative PCNA expression
NC OE KD
J
Scaled expression
Min Max
NC POU2F1 -OE
GO terms
Endochondral bone morphogenesis (FOXC1, SCX)
Regulation of cyclin−dependent protein kinase activity (PCNA, CDK1, CDKN1C, CDKN2C, TTBK1)
Regulation of integrin activation (PTGER4, PLEK)
Chemokine−mediated signaling pathway (FOXC1, CCL23, CCL5, PTK2B)
POU2F1 -OE up
log10(pvalue)
DNA metabolic process (PRIM2, STON1, PDS5B, BACH1)
Regulation of mitochondrial mRNA stability (FASTKD1, FASTKD2, FASTKD3)
Long−chain fatty−acyl−coA metabolic process (HACD1, ACSL1, ACSL4, ACSBG1)
Sphingolipid metabolic process (UGCG, ARV1, HACD1, ACER3)
POU2F1 -OE down
log10(pvalue)
K
Down-regulated in POU2F1 -KD Hep & up-regulated in Hepa2
with POU2F1 binding motif (-2000 bp ~ +500 bp)
449
160
L
ERBB signaling
Cell cycle
Positive regulation of developmental process
WNT signaling
GAB1 KIF16B PTK2 ERBIN NRG1 IQGAP1 UBR5 DEPDC1B TBL1XR1 TNKS LGR4
NCAPD3 CDC14A CIT GPSM2 ECT2 KIF20B RAD21 RRM1 CDK1 NUSAP1 PKP4 BRCA1 E2F7 TICRR RBBP8 DBF4 NCAPG RAD18 MRE11 CHEK1 BLM BARD1
SHLD2 NCOA2 CLASP1 RBBP7 SS18 ARID2 ARID1B OGT BMPR1A
M
Frequency
-3k 0 3k
Position around motif (bp)
NC POU2F1 -OE
N
NC
POU2F1-OE
PTPN3
KCNT2
LGR4
TBL1XR1
Cell cycle
WNT pathway
O
Hep
Pro-hep
KC
Endo
Cho
T cell
HSC
Portal vein
Cental vein
Regenerating hepatocytes (POU2F1high)
Cytoplasm
POU2F1
Promoter
Nucleus
WNT target genes (LGR4, LGR5, TBL1XR1...)
Cell cycle genes (CDC14A, KCNT2, PTPN3...)
...